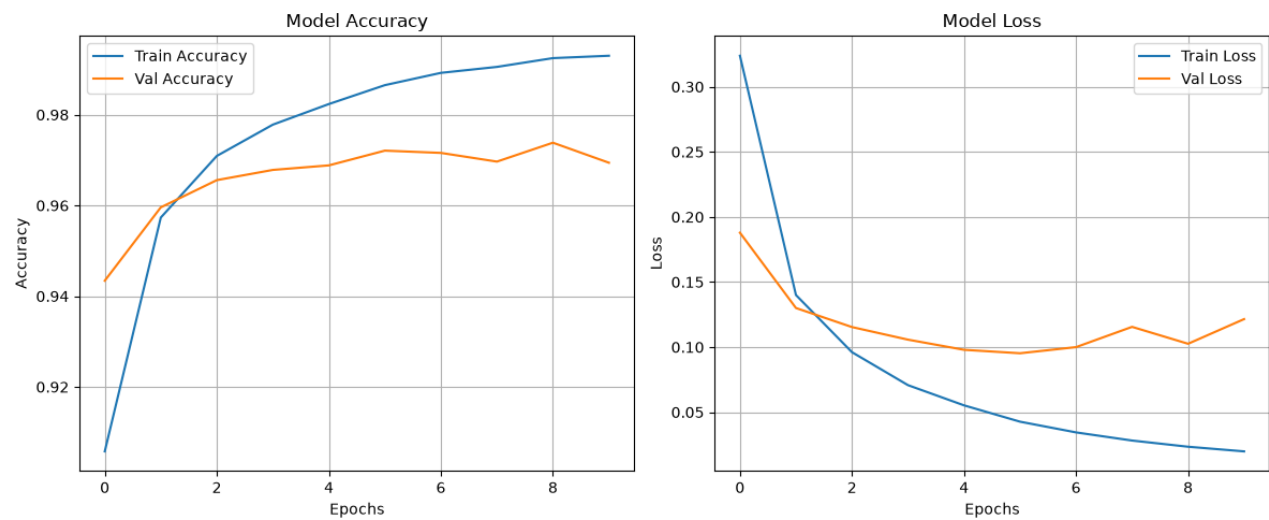
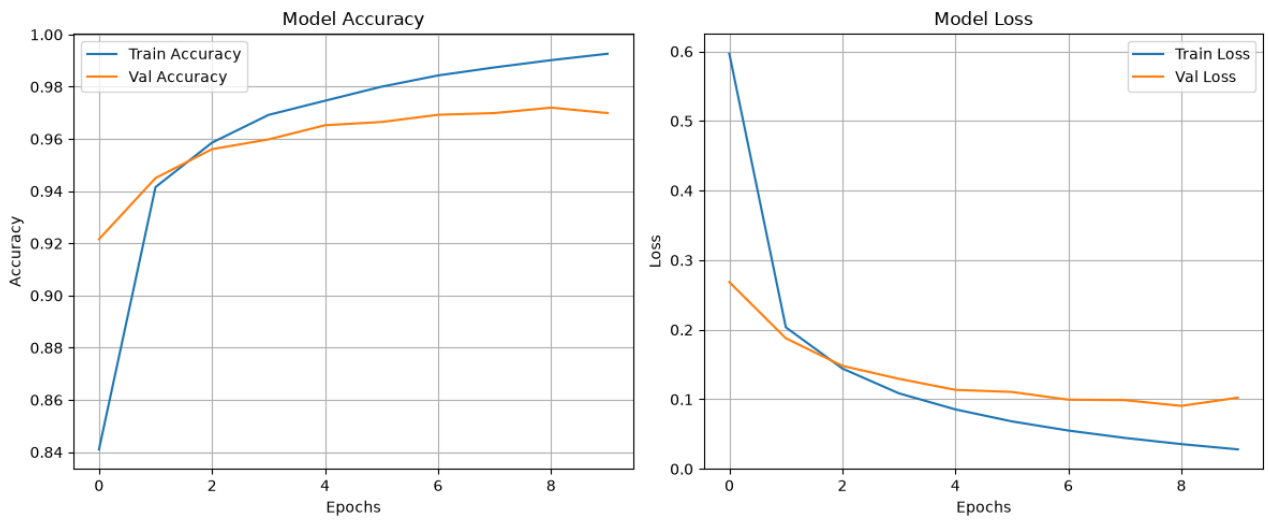


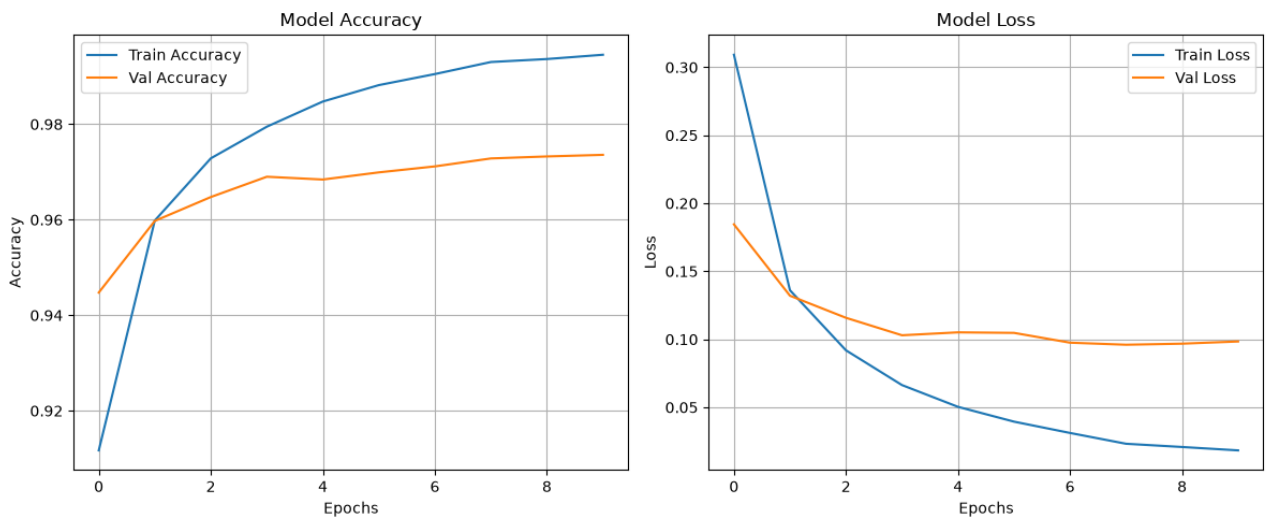
Layers (128, 64), Act: ReLU, No Drop-Out



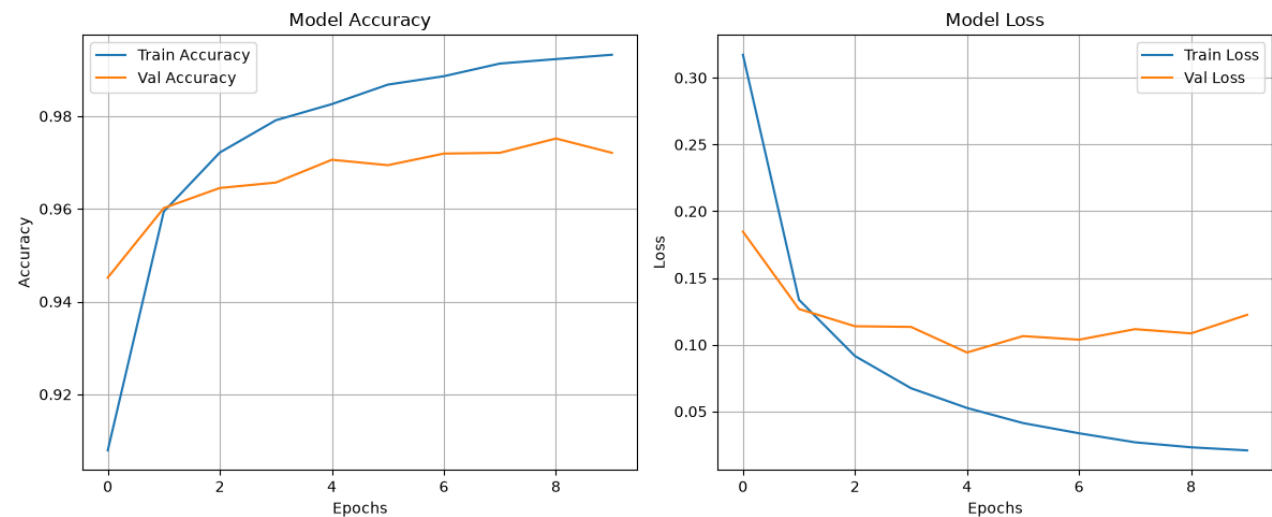
Layers (128, 64), Act: **Sigmoid**, No Drop-Out



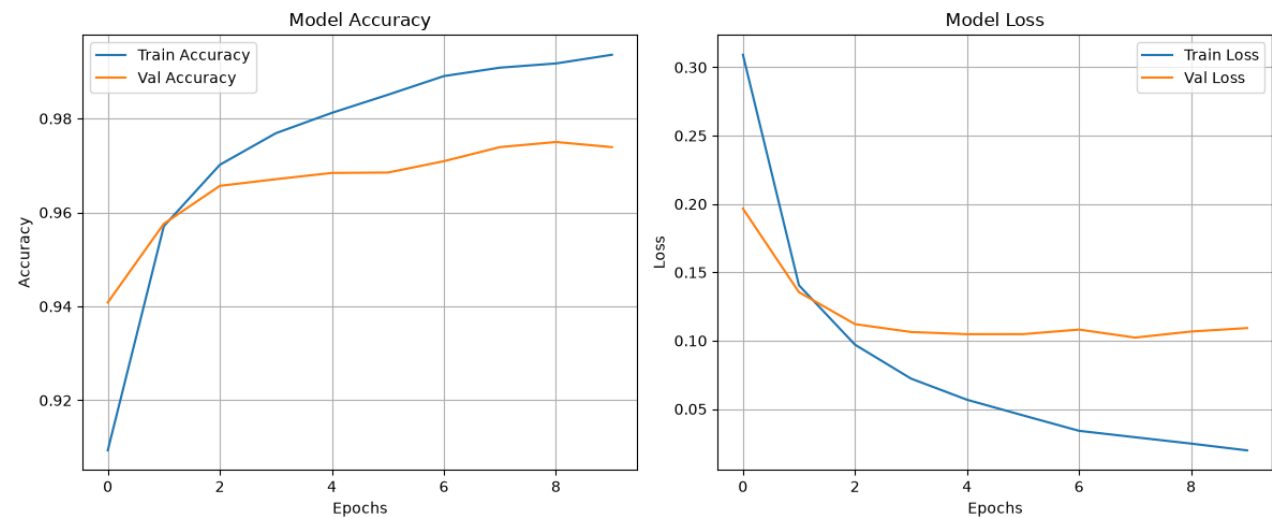
Layers (128, 64), Act: **Tanh**, No Drop-Out



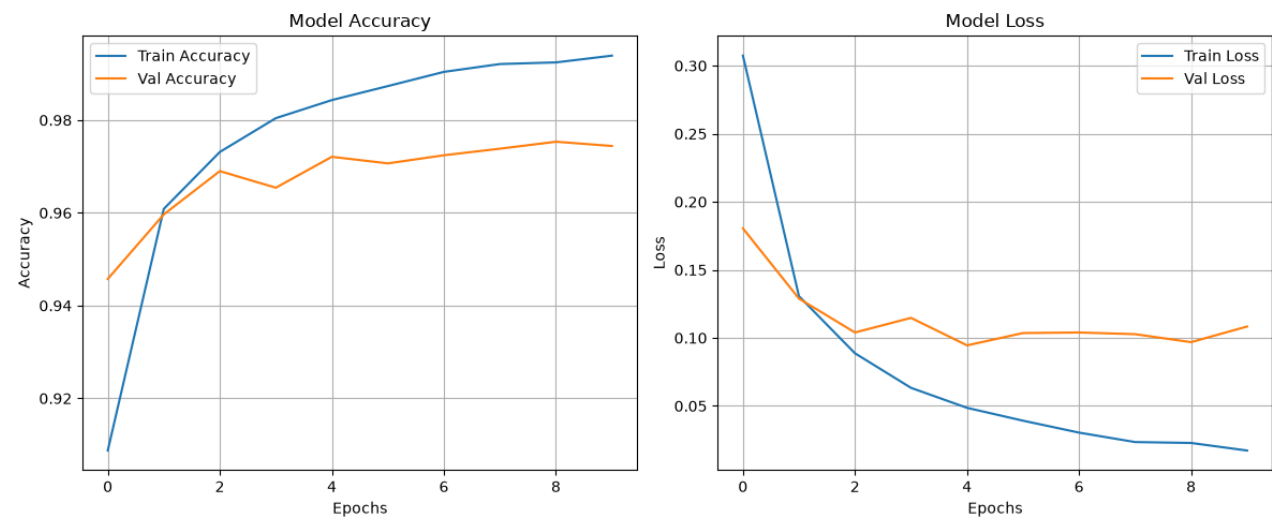
Layers (128, 64), Act: **LeakyReLU**, No Drop-Out



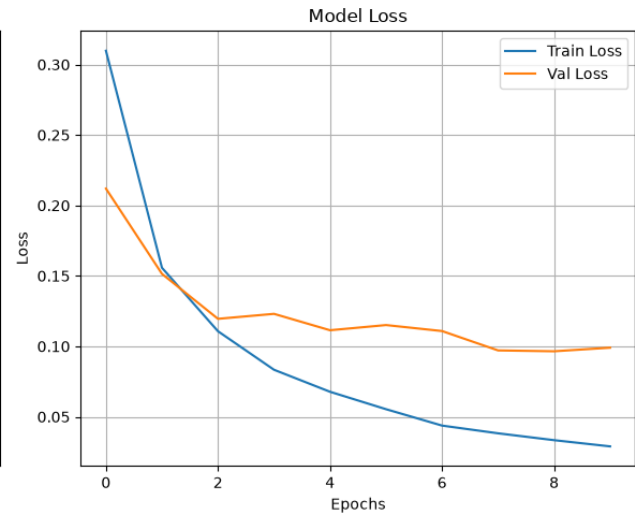
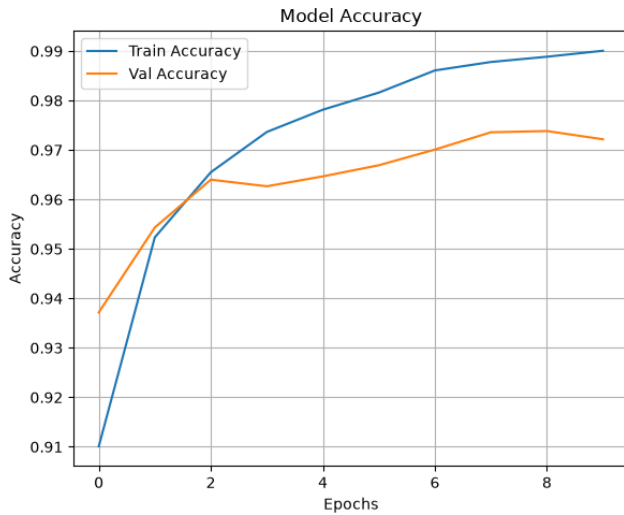
Layers (128, 64), Act: **ELU**, No Drop-Out



Layers (128, 64), Act: **GELU**, No Drop-Out



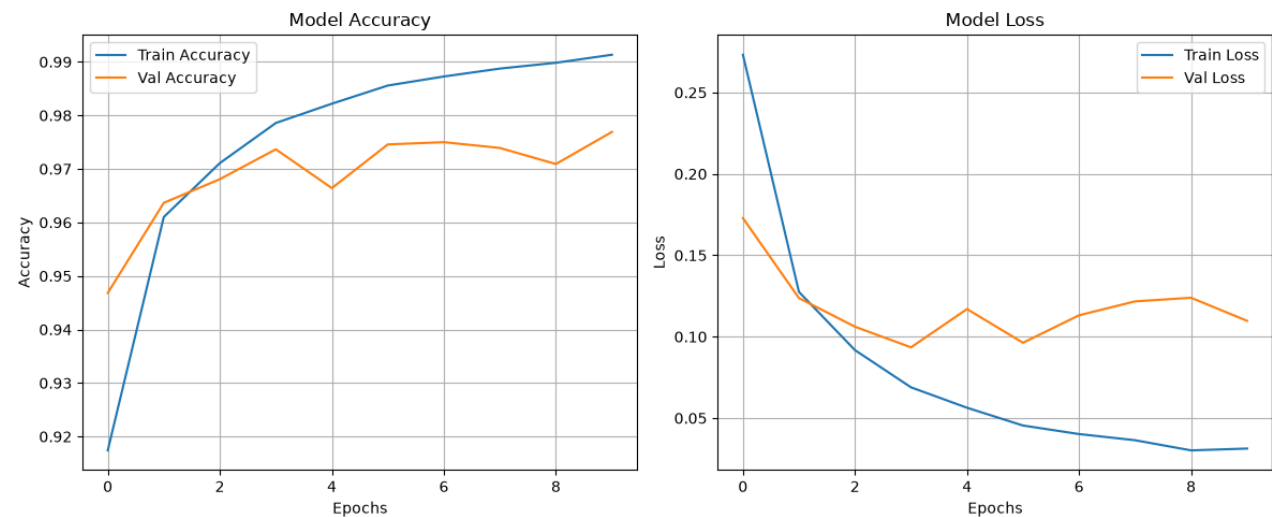
Layers (128, 64), Act: **SELU**, No Drop-Out



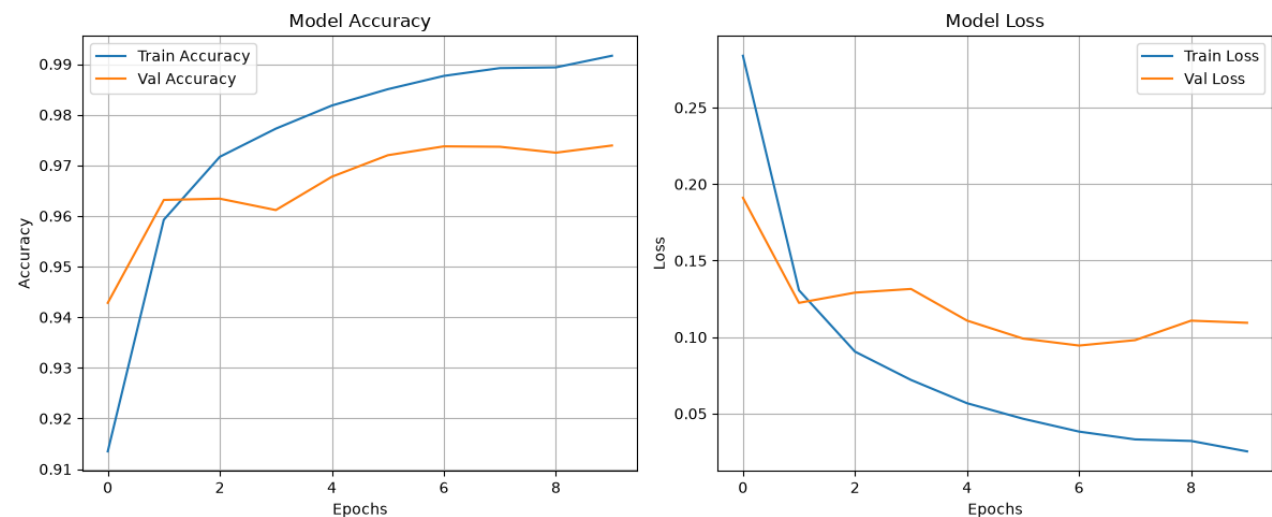
Layers (128, 64), No Drop-Out, No Stop Loss

Activation	Epoch 1 Val Acc	Epoch 10 Val Acc	Final Test Accuracy	Epoch 1 Val Loss
ReLU	0.9434	0.9694	0.9718	0.1879
Sigmoid	0.9216	0.9699	0.9735	0.2685
Tanh	0.9447	0.9736	0.9756	0.1845
LeakyReLU	0.9452	0.9721	0.9757	0.1847
ELU	0.9408	0.9739	0.9777	0.1966
GELU	0.9457	0.9744	0.9763	0.1805
SELU	0.9372	0.9722	0.9738	0.2120

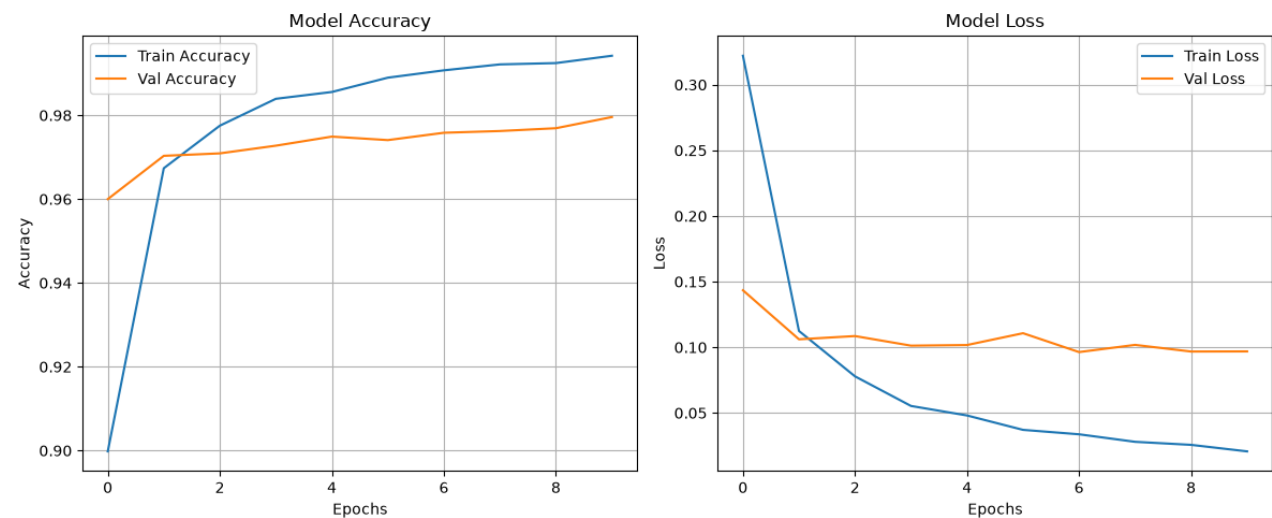
Layers **(512, 256)**, Act: **ELU**, No Drop-Out



Layers **(512, 128, 32)**, Act: **ELU**, No Drop-Out



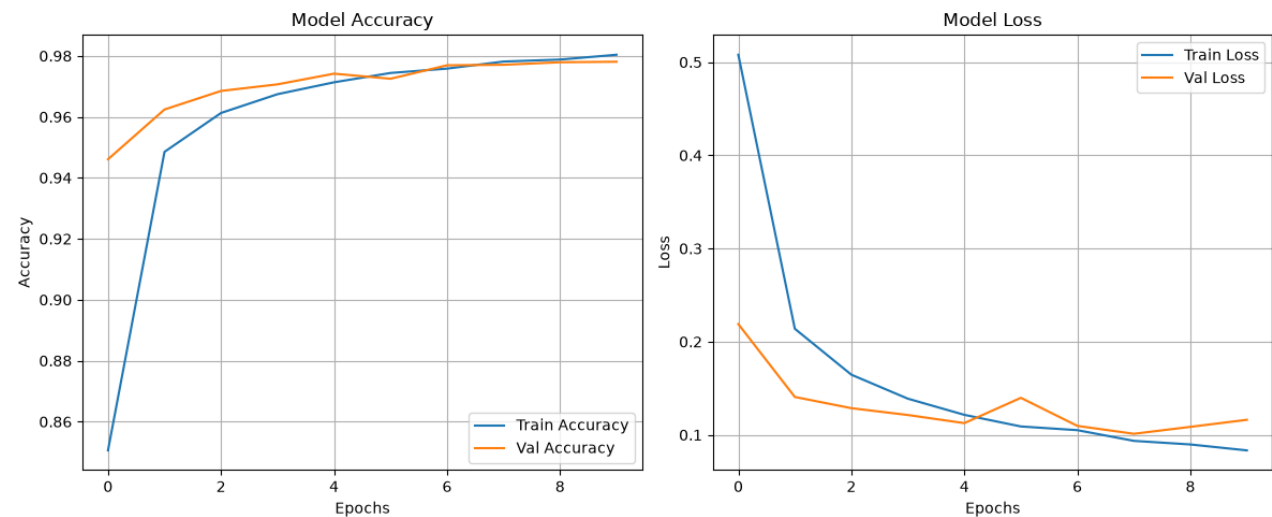
Layers **(512, 256, 128, 64, 32)**, Act: **GELU**, No Drop-Out



Batch: 32, No Drop-Out

Architecture	Activations	Epoch 1 Val Acc	Final Train Acc	Final Val Acc	Test Accuracy
256 → 32	ELU/ELU	0.9447	0.9942	0.9752	0.9747
128 → 128	ELU/ELU	0.9376	0.9928	0.9742	0.9754
512 → 256	ELU/ELU	0.9468	0.9913	0.9769	0.9787
256 → 128 → 64	ELU/ELU/ELU	0.9430	0.9907	0.9701	0.9735
256 → 128 → 64 (rerun)	ELU/ELU/ELU	0.9514	0.9914	0.9758	0.9763
512 → 128 → 32	ELU/ELU/ELU	0.9428	0.9916	0.9739	0.9736
512 → 128 → 32	ELU/GELU/Sigmoid	0.9419	0.9914	0.9760	0.9760
128 → 128 → 32	ELU/GELU/Sigmoid	0.9413	0.9937	0.9708	0.9718
32 → 32 → 32	ELU/GELU/Sigmoid	0.9084	0.9791	0.9642	0.9673
512 → 256 → 64	ELU/GELU/Sigmoid	0.9438	0.9919	0.9755	0.9785
512 → 256 → 64	ELU/GELU/GELU	0.9417	0.9905	0.9724	0.9737
512 → 256 → 64	Sigmoid×3	0.9318	0.9939	0.9702	0.9735
512 → 256 → 128 → 6 4 → 32	ELU×5	0.9442	0.9905	0.9777	0.9801
512 → 256 → 128 → 6 4 → 32	ELU/GELU/ELU/ GELU/Sigmoid	0.9337	0.9832	0.9702	0.9735
512 → 256 → 128 → 6 4 → 32	GELU×5	0.9600	0.9942	0.9796	0.9808

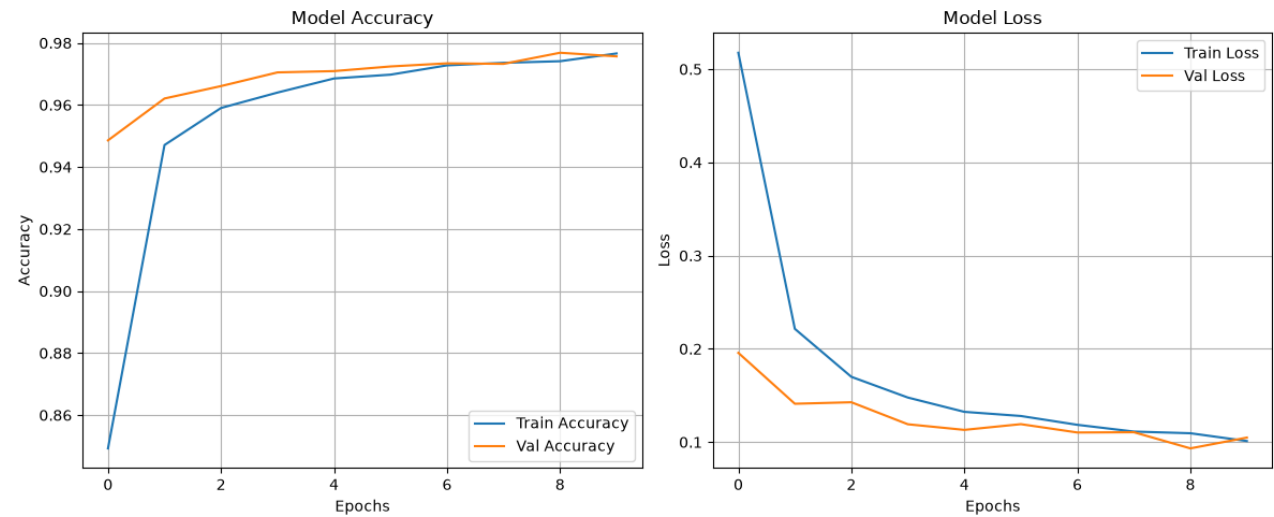
Layers (512, 256, 128, 64, 32), Act: GELU, Drop-Out: 0.3



Batch: 32, Hidden Layers (512, 256, 128, 64, 32), Act: GELU

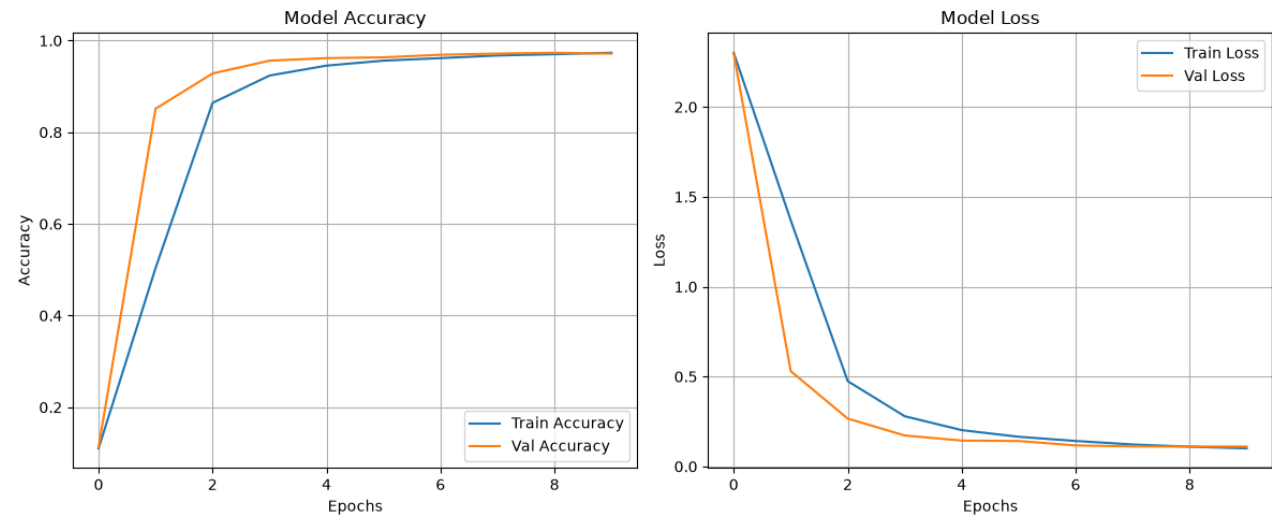
Dropout Rate	Last Activation	Final Train Acc	Final Val Acc	Test Accuracy
0.2	GELU	0.9874	0.9798	0.9808
0.3	GELU	0.9804	0.9781	0.9787
0.3	Sigmoid	0.9769	0.9761	0.9790
0.35	GELU	0.9755	0.9722	0.9755
0.4	Sigmoid	0.9643	0.9740	0.9736
0.4	GELU	0.9718	0.9743	0.9748
0.5	GELU	0.9459	0.9730	0.9744

Layers (512, 256, 128, 64, 32), Act: GELU, Drop-Out: 0.3, L2 Reg: 1e-4



Dropout	L2 (weight_decay)	Train Acc (final)	Val Acc (final)	Test Accuracy
0.0	none	0.9942	0.9796	0.9808
0.0	1e-4	0.9889	0.9739	0.9792
0.2	none	0.9874	0.9798	0.9808
0.2	1e-4	0.9816	0.9735	0.9754
0.3	none	0.9804	0.9781	0.9787
0.3	1e-4	0.9766	0.9757	0.9769
0.5	none	0.9459	0.9730	0.9744
0.5	1e-4	0.9459	0.9707	0.9709

Layers (512, 256, 128, 64, 32), Act: GELU, Drop-Out: 0.3, L2 Reg: 1e-4, Opt: SGD



Layers (512, 256, 128, 64, 32), Act: GELU, Drop-Out: 0.3, L2 Reg: 1e-4, Opt: RMSprop

